

BALLISTOCARDIOGRAM-BASED HEART RATE VARIABILITY ESTIMATION FOR STRESS MONITORING USING CONSUMER EARBUDS

David J. Lin¹, Md Mahbubur Rahman², Li Zhu², Viswam Nathan², Jungmok Bae², Christina Rosa³, Wendy B. Mendes³, Jilong Kuang², Alex J. Gao²

¹Georgia Institute of Technology, ²Samsung Research America, ³University of California San Francisco

ABSTRACT

Stress can potentially have detrimental effects on both physical and mental well-being, but monitoring it can be challenging, especially in free-living conditions. One approach to address this challenge is to use earbud accelerometers to capture the ballistocardiogram (BCG) response. These sensors allow for noninvasive stress monitoring by estimating physiological indicators linked to stress, such as heart rate variability (HRV). However, ear-worn devices are susceptible to motion artifacts and can exhibit significant BCG signal morphology variations. These challenges necessitate accurate algorithms to estimate HRV for everyday use. Therefore, we developed a method to measure interbeat intervals (IBI) from BCG signals collected from an earbud. To enhance IBI estimation accuracy, we employed a Bayesian method that incorporates robust apriori IBI prediction weighting and sensor fusion techniques. We have also conducted a study involving 97 participants to assess the earbuds' ability to estimate HRV metrics and classify stressful activities. Our findings demonstrate low IBI estimation error ($4.16\% \pm 1.90\%$), along with lower errors in subsequent higher-order HRV metrics compared to the state-of-the-art algorithms.

Index Terms— Ballistocardiogram (BCG), Heart Rate Variability (HRV), Stress Detection, Wearables

1. INTRODUCTION

Stress remains a persistent concern, with the potential to worsen both mental and physical health problems, highlighting the importance of passive, continuous stress monitoring systems for individuals to proactively manage it. One method of identifying stress involves the continuous monitoring of stress-related biomarkers such as heart rate variability (HRV) or interbeat intervals (IBI) variations in time and frequency domains [1]. These biomarkers can be noninvasively measured and offer a convenient means to detect stress before it escalates.

The ballistocardiogram (BCG), a noninvasively acquired cardiomechanical signal measuring blood ejection recoil, is of

particular interest for HRV monitoring because of its sensing convenience. Unlike the electrocardiogram (ECG), the gold-standard for HRV measurements [2], which require cumbersome electrode placement [3], the BCG can be conveniently measured using inertial measurement units (IMUs). Typically BCG sensors have been integrated into large immobile products such as beds or chairs [4, 5, 6] that hinder free-living monitoring. However, the versatility and portability of IMUs enable BCG sensing in more mobile and lightweight devices. As smartwatches and earbuds with IMUs become increasingly prevalent, these existing sensors can be repurposed for BCG sensing, enabling continuous daily HRV monitoring.

The ear, in particular, is a promising sensing location for mobile monitoring as earbuds are small and widely used by consumers. The ear BCG signal has been explored in previous works [7, 8], though not for HRV estimation, and holds some advantages over noninvasive photoplethymogram (PPG) sensing [9]. While there have been some commercially available PPG earbuds [10] or more commonly, PPG smartwatches, they typically require specialized hardware that increases the size and power consumption of these devices. In contrast, IMUs commonly integrated into these earbuds can be easily adapted for cost-effective BCG monitoring. Furthermore, earbud BCG offers superior signal fidelity and motion artifact tolerance compared to BCG on the wrist attributed to the higher degrees of freedom at the arm.

Nevertheless, employing the ear BCG for IBI and HRV estimation presents challenges due to its susceptibility to motion artifacts (MA) and its quality being contingent on the fit of the earbud. Motion artifacts can introduce substantial errors in IBI measurements, subsequently resulting in larger errors in HRV metrics [11]. While previous studies have explored IBI detection using non-ear BCG signals with minimal errors, these investigations were predominantly confined to scenarios with high signal-to-noise ratios (SNR), such as during sleep or when lying down [12, 13]. These findings may be more challenging to apply in low SNR, real-world settings.

In this work, we present a pipeline depicted in Figure 1 that enables HRV estimations using an earbud accelerometer. We improved HRV measurement accuracy in low data yield environments by firstly, fusing BCG signal information be-

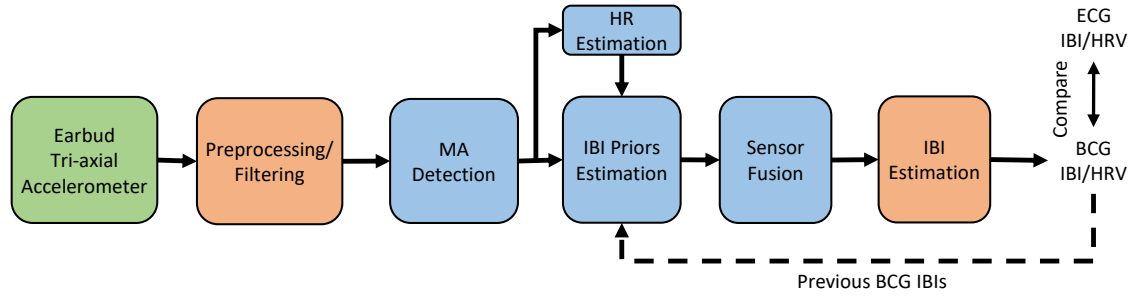


Fig. 1. Interbeat interval (IBI) and heart rate variability (HRV) estimation pipeline using an earbud. BCG signals acquired using a tri-axial accelerometer are filtered and motion artifacts (MA) are removed. IBIs are then estimated and probable IBIs are further enhanced by using a weighted IBI prior and fused information from different axes. This prior is constructed using previous IBI estimates and an independent HR estimator. The resulting parameters are compared with reference ECG IBI and HRV parameters for evaluation.

tween different axes to also accommodate device orientation fitting variations and secondly, utilizing robust IBI priors to reduce high IBI estimation variation and more accurately determine the current IBI even after high noise data. We also tested the algorithm's performance on a large population set with 97 subjects going through various stressors and relaxations. Finally, we tested the earbud's capability to detect stress events using the derived HRV metrics. These contributions suggest that earbud-based BCG sensing can be used for physiological monitoring and detecting minute changes in IBI and HRV in response to stress.

2. METHODS

2.1. Data collection

We recruited 97 subjects (Age: 18-51) and collected earbud data and reference physiological signals simultaneously. During the protocol, subjects were underwent various tasks, including baseline rest, public speaking and preparation, mental arithmetic, music listening, and cold immersion, each 4 minutes long. This protocol was approved by the University of California San Francisco Institutional Review Board (IRB).

BCG signals were collected concurrently during these tasks using a dual earbud sensor. The right earbud contained a tri-axial accelerometer sampled at 50Hz, while the left earbud contained a green wavelength PPG and a tri-axial accelerometer sampled at 25Hz. In this work, we only used the higher sampled 50Hz accelerometer signal for its higher BCG signal fidelity and its measurement convenience using IMUs already integrated into commercial earbuds. ECG was concurrently collected using a Biopac data acquisition system at 1kHz. The ECG was then annotated by experts and associated HRV metrics were extracted, including root mean square of successive differences (RMSSD), the standard deviation of normal to normal (NN) heartbeat intervals (SDNN), the percentage of NN intervals above 50ms (PNN50), and respiratory sinus

arrhythmia (RSA), a commonly used frequency-based HRV.

2.2. Preprocessing and motion artifact (MA) detection

The BCG signal was first resampled to 50Hz and filtered using a finite impulse response (FIR) zero-phase Butterworth filter with cutoffs of 1-12Hz to remove breathing artifacts while preserving BCG features [13]. A Savitzky-Golay filter with a window length of 1 second was used to remove any baseline wanders or shifts due to earbud fit adjustments [14].

To prevent motion-related signal interference in IBI estimation, we excluded segments that had too large of a magnitude response from all three axes [15]. We set a threshold based on subjects' baseline periods and only evaluated segments that had 10 seconds of uninterrupted good quality to ensure an adequate number of windows for analysis. With this approach, we prioritized IBI accuracy over uninterrupted estimation as HRV relies on accurate IBI estimates and can still be estimated on data even with some windows removed.

2.3. Interbeat interval (IBI) estimation

To estimate IBI, we adopted an algorithm that uses a Bayesian approach and speech processing-inspired techniques to estimate IBI on a window-by-window basis [12]. This algorithm was chosen for its interpretability and low computational overhead for future on-board deployment and has been employed in recent works [13, 16]. In this algorithm, three periodicity estimators were evaluated on each window across a range of IBI candidates. Each estimator's output was normalized to form a probability density function (PDF) of the likelihood of each IBI candidate. These PDFs were then multiplied together to form a more robust PDF. The IBI candidate with the highest PDF value was chosen as the IBI estimate. The associated PDF value was also used as a quality metric as low-quality valued BCG signals had low periodic behavior. Using this quality metric, unreliable IBI values with low-

Table 1. HRV metric errors across subjects between BCG and ECG. Configurations tested: **Base**, the pipeline without IBI apriori weighting and sensor fusion, **Single**, the pipeline without sensor fusion, and **Proposed**, the pipeline with all components. Mean absolute error (MAE), Mean absolute percentage error (MAPE), and Symmetric MAPE (SMAPE) were calculated.

Features	Base			Single			Proposed		
	MAE	MAPE [%]	SMAPE [%]	MAE	MAPE [%]	SMAPE [%]	MAE	MAPE [%]	SMAPE [%]
IBI [ms]	79.0	10.0	4.92	54.4	6.58	3.38	34.5 ± 16.6	4.16 ± 1.90	2.07 ± 0.99
RMSSD [ms]	88.4	197.5	36.1	26.6	51.7	18.8	21.6 ± 9.11	35.3 ± 22.1	19.3 ± 7.16
SDNN [ms]	70.5	246.2	38.0	32.1	95.4	24.6	19.7 ± 11.5	56.7 ± 89.7	17.4 ± 7.66
RSA	1.77	31.2	11.9	0.88	15.4	6.61	0.61 ± 0.29	10.5 ± 8.70	4.83 ± 2.70
LF/HF Ratio	1.57	95.7	44.6	1.56	103	38.6	1.43 ± 1.25	87.6 ± 47.1	35.0 ± 7.82
PNN50 [%]	20.2	131.3	148.9	14.3	94.2	31.0	8.72 ± 3.83	41.4 ± 28.8	29.4 ± 18.2

quality values were removed.

In this work, IBI candidates ranged from 300 to 1500ms. Window length and overlap were set to 3s and 0.2s respectively to ensure at least 2 beats would always be in a window. The quality threshold was set at 0.15.

2.3.1. IBI Weighting

Because IBI is estimated independently for each window in [12], noise can cause egregious IBI estimates. Motivated by IBI-to-IBI stability, we weighted the current window's PDF with a prior IBI distribution, as similarly done by [13]. This prior is a Gaussian distribution centered around a predicted IBI value, with a standard deviation of 0.1 to match ECG-based IBI changes. Additionally, we further improved the predicted IBI estimate by averaging two predictions: one from the median of the algorithm's previous N IBI estimates encompassing 10 seconds and one from an independent heart rate (HR) estimate over a larger window.

The HR-estimator was used because artifacts can heavily bias the prior when only using previous IBI estimates as the periodicity functions were designed to use as few samples as possible to estimate IBI. Moreover, with a short look-back period, re-initializing to the correct IBI after large data gaps can be challenging in low SNR scenarios. To overcome this, we enhance the IBI prediction with an independent HR-based IBI estimator, utilizing the close relationship between HR and IBI and the robustness of longer computing windows. We apply principles of PPG HR estimation on the BCG by tracking the HR frequency on its spectrogram [17] with an 8-second window and 80% overlap. Before calculating the spectrogram, the Hilbert transform in the BCG is used to accentuate HR fundamental frequencies [18]. HR is then calculated as the maximum along the 0.7-4Hz range (to match the IBI candidate range) and tracked using a single-state Kalman filter to mitigate unreliable estimates. The state was BCG HR with dynamics set such that HR was presumed constant. [8].

2.3.2. Sensor Fusion

Due to factors like earbud orientation and subject-specific parameters, BCG components exist in multiple accelerometer axes. While the BCG signal was more apparent for most subjects in the Y-axis (anterior posterior) of the sensor, the X-axis (superior inferior) sometimes had an equally apparent BCG signal. To utilize the shared BCG structure, the PDF was calculated for each axis and averaged together to form the final PDF. This approach was inspired by other multichannel-based work; however, we use the axes as the different channels instead of separate spatially distant sensors [19, 20]. This fusion enhances common periodic BCG components while reducing uncorrelated noise across axes.

2.4. Algorithm Evaluation

To evaluate the algorithm, we calculated the mean absolute error (MAE), mean absolute percentage error (MAPE), and symmetric MAPE (SMAPE) of the estimated BCG IBI to the ECG IBI. This was also repeated for RMSSD, SDNN, PNN50, and RSA.

Metrics were calculated across all patients for different algorithm configurations – Base, Single, and Proposed. Base refers to the base algorithm developed by [12] while the Single configuration adds the modified IBI prior functions. In both configurations, the Y-axis of the earbud accelerometer was used as it commonly contained the strongest BCG component. Finally, Proposed adds in the sensor fusion module, incorporating the different accelerometer axes. This analysis helps highlight the benefits of each addition.

We also compared the potential of using BCG vs ECG-derived HRV metrics to classify stress and nonstress tasks. This comparison aimed to gauge whether BCG-derived HRV errors were significant enough to impact downstream tasks. Stress tasks were public speaking and prep, and mental arithmetic. Nonstress tasks were baseline and music listening.

We used a random forest (RF) classifier with all available features, applying a 5-fold cross-validation, separating

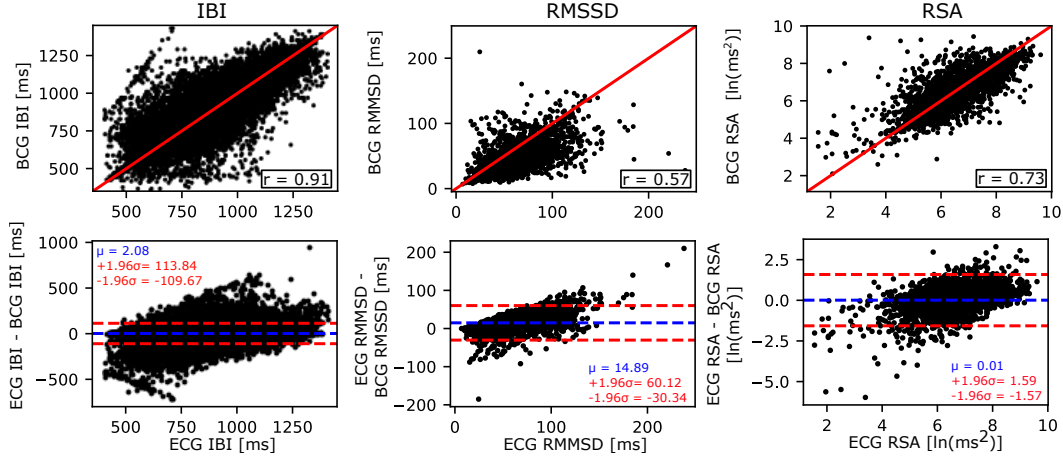


Fig. 2. Correlation (top row) and Bland Altman (bottom row) plots for Earbud based BCG against reference ECG IBI, RMSSD, SDNN, and RSA. The red line in the correlation plot denotes the identity line. The dotted blue and red lines in the Bland Altman plots show the average difference and limits of agreement respectively.

on subjects. Input data was normalized using their baseline measurements. Performance was measured using balanced accuracy, F1 score, recall, and precision. RF parameters were set at 100 trees and a max depth of 6 to prevent overfitting.

3. RESULTS

Table 1 summarizes algorithm performance for all configurations. The Base configuration exhibits notably high IBI MAPE leading to increased higher HRV errors. In contrast, the Single configuration with IBI apriori weighting showed considerable improvement in IBI estimation. The weighting applied enabled a better prediction of where the next IBI should be, resulting in a lower error in all HRV metrics. Finally, the Proposed configuration, which combines accelerometer axes' information, results in the lowest IBI and HRV errors across all configurations. It achieved an IBI MAPE less than 5%, greatly improving HRV accuracy, as evidenced by the strong agreement to the ECG reference in Figure 2. This suggests that other axes contain pertinent BCG content.

In comparison to other studies [12, 13], the IBI MAE might appear high. However, considering the 50Hz sampling rate of the earbud, the 34.5ms MAE corresponds to less than a 2-sample difference to the 1kHz sampled ECG-derived IBI. Additionally, these works focused on stationary BCG devices and low-motion activities. Future work may explore advanced sensor fusion techniques, including the best channel selection or weighted fusion based on signal quality.

HRV metrics were less accurate than IBI estimation. This accuracy drop is due to firstly, slight errors in IBI values propagating into larger HRV metric errors when examining successive IBI values and secondly, lower coverage of BCG IBI values due to motion artifacts resulting in fewer beats being

Table 2. Stress Classification using different modalities

Signal	Precision	Recall	Accuracy	F1
BCG	0.83	0.62	0.77	0.70
ECG	0.77	0.62	0.77	0.69

used for HRV estimation [11].

3.1. Stress Classification

Table 2 shows the results of stress classification using HRV features derived from either the ECG or BCG. The similar performance and consistent feature importances between the two modalities suggests that the estimation errors do not affect classification performance. The low overall F1 score can be attributed to the low data yield during higher motion artifact afflicted tasks like public speaking and arithmetic tasks. Because these tasks are more stress-inducing than public speaking prep, the lack of these task samples lowers the recall score. This was confirmed when classifying stress using all reference ECG HRV metrics, resulting in an increased F1 score of 0.83. Further denoising of the signal may increase data yield during these noisier sections.

4. CONCLUSION

We demonstrate IBI estimation using an earbud accelerometer. By using robust IBI priors and combining BCG components from different axes, we achieved high IBI and HRV accuracy. Moreover, HRV estimation error had minimal effect on stress classification performance. Through this, we showed the potential of using consumer earbuds for unobtrusive HRV and stress monitoring during daily activities.

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